IMAGING COURSE · LESSON 14

Thyroid and salivary glands

The thyroid, parotid and submandibular glands lie close to the skin, so **ultrasound** is the first test for almost every lump in them, ideally with a needle in the same sitting. CT and MRI come next, for what ultrasound cannot see: the deep lobe of the parotid, the retrosternal goitre, invasion of the airway, perineural spread and the nodes behind the pharynx. This lesson covers the normal glands, the ultrasound features that grade risk, and the cross-sectional signs that change an operation.

AT A GLANCE

- **Ultrasound first**, with FNA, for thyroid and salivary lumps.
- Suspicious nodule: **solid, hypoechoic, taller than wide, irregular, microcalcifications**. Grade by **U** or **TI-RADS**.
- **CT** for retrosternal goitre and invasive cancer; remember iodine delays radioactive iodine.
- The facial nerve lies **lateral to the retromandibular vein**; deep lobe tumours widen the **stylomandibular gap**.
- Pleomorphic adenoma: very **bright on T2**. Warthin: parotid **tail**, multiple, smokers. Low T2 and ill-defined margins suggest malignancy.

1 CHAPTER ONE The thyroid

Ultrasound first.

1.1 The normal gland and its relations

The thyroid wraps the front of the trachea from the level of the cricoid down, with the isthmus across the second to fourth tracheal rings. On ultrasound it is homogeneous and brighter than the strap muscles; on CT it is dense even before contrast because of its iodine, and enhances avidly. The **carotid** and **jugular vein** lie laterally, the **oesophagus** behind the left lobe, and the **tracheoesophageal grooves** hold the recurrent laryngeal nerves, the paratracheal (level VI) nodes and the parathyroid glands.

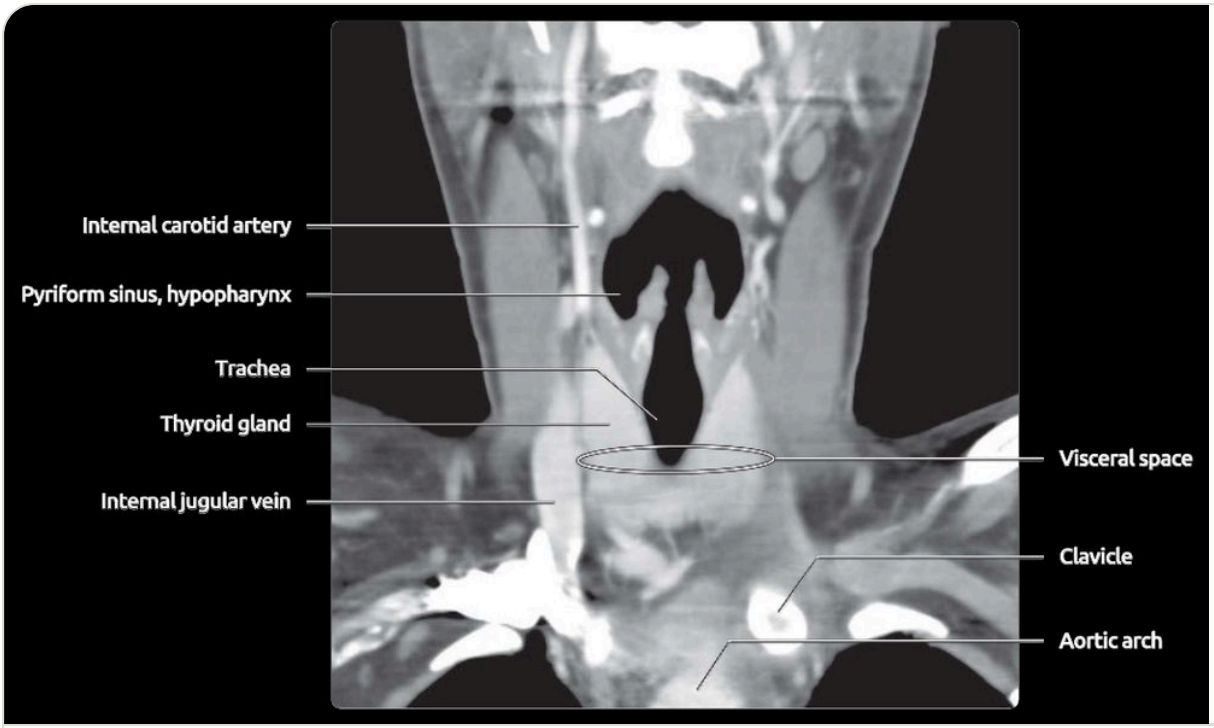


FIGURE 1 Coronal contrast CT: the chevron-shaped thyroid in the visceral space, wrapping the trachea below the larynx.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 369

1.2 Grading a nodule

Ultrasound grades a nodule's risk by its features. The UK **U grading** (U1 normal, U2 benign, U3 indeterminate, U4 suspicious, U5 malignant) and the American **TI-RADS** (TR1 to TR5, scored by points) use the same features:

ULTRASOUND FEATURES OF THYROID NODULES

FEATURE	REASSURING	SUSPICIOUS
Composition	Cystic, spongiform	Solid
Echogenicity	Iso- or hyperechoic, with a halo	Hypoechoic, especially markedly hypoechoic
Shape	Wider than tall	**Taller than wide** on transverse view
Margin	Smooth, well defined	Lobulated, irregular, extrathyroidal extension
Echogenic foci	Comet-tail artefact (colloid), eggshell rim	**Microcalcifications**, interrupted rim calcification

FEATURE	REASSURING	SUSPICIOUS
Nodes	Normal	Round, cystic or calcified nodes in levels III, IV, VI

Under the British Thyroid Association guidance, U3 to U5 nodules are sampled by FNA; under ACR TI-RADS the size threshold for FNA falls as the score rises (for example, 1 cm for TR5 and 2.5 cm for TR3). Cytology is reported in the Thy (UK) or Bethesda categories.



FIGURE 2 Differentiated thyroid carcinoma on ultrasound: a well-defined nodule with many tiny hyperechoic microcalcifications. FNA showed papillary carcinoma.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 238

1.3 When to add CT or MRI

Ultrasound cannot see behind the sternum or the trachea. **CT** (or MRI) is added for a **retrosternal goitre** (its lowest extent, tracheal narrowing and deviation, and whether it can be delivered through the neck), for a cancer that may invade the **trachea, larynx, oesophagus or recurrent nerve**, and for nodes in the mediastinum or behind the pharynx. In differentiated thyroid cancer, iodinated contrast delays radioactive iodine treatment by weeks, so the need for contrast is discussed with the team; MRI avoids the problem.



FIGURE 3 Multinodular goitre. Left: a chest film, the trachea displaced by a mass in the neck and upper mediastinum. Right: coronal contrast CT, heterogeneous enlargement of both lobes, the left extending into the mediastinum.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 226

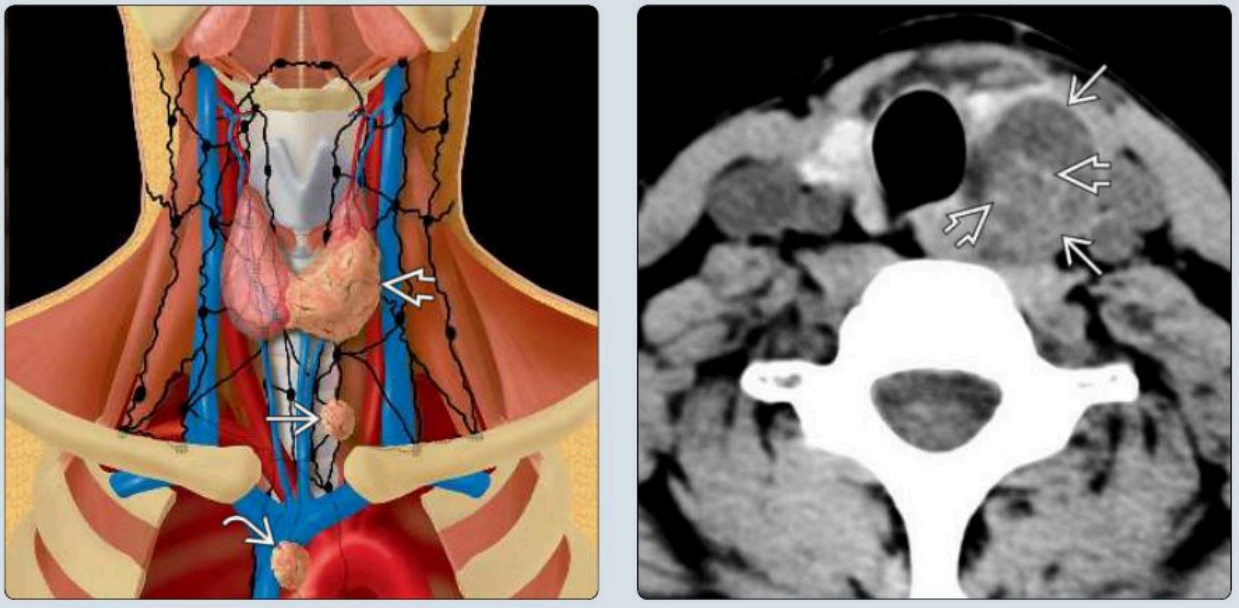


FIGURE 4 Differentiated thyroid carcinoma on CT. Left: drawing of a tumour in the left lobe. Right: non-contrast CT, a well-defined mass in the left lobe with fine, speckled microcalcifications.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 238

1.4 The incidental nodule and the parathyroids

Thyroid nodules are often found by chance on CT, MRI or PET. Uptake of FDG in a focal nodule carries a significant risk of cancer and needs ultrasound; a small, non-

avid nodule on CT in an older patient usually does not. For **primary hyperparathyroidism**, ultrasound and **sestamibi** scintigraphy (often with SPECT-CT) localise most adenomas, and **four-dimensional CT** helps when they disagree or after failed surgery. An adenoma is a small, oval, hypoechoic nodule behind the thyroid, often with a feeding vessel at one pole.

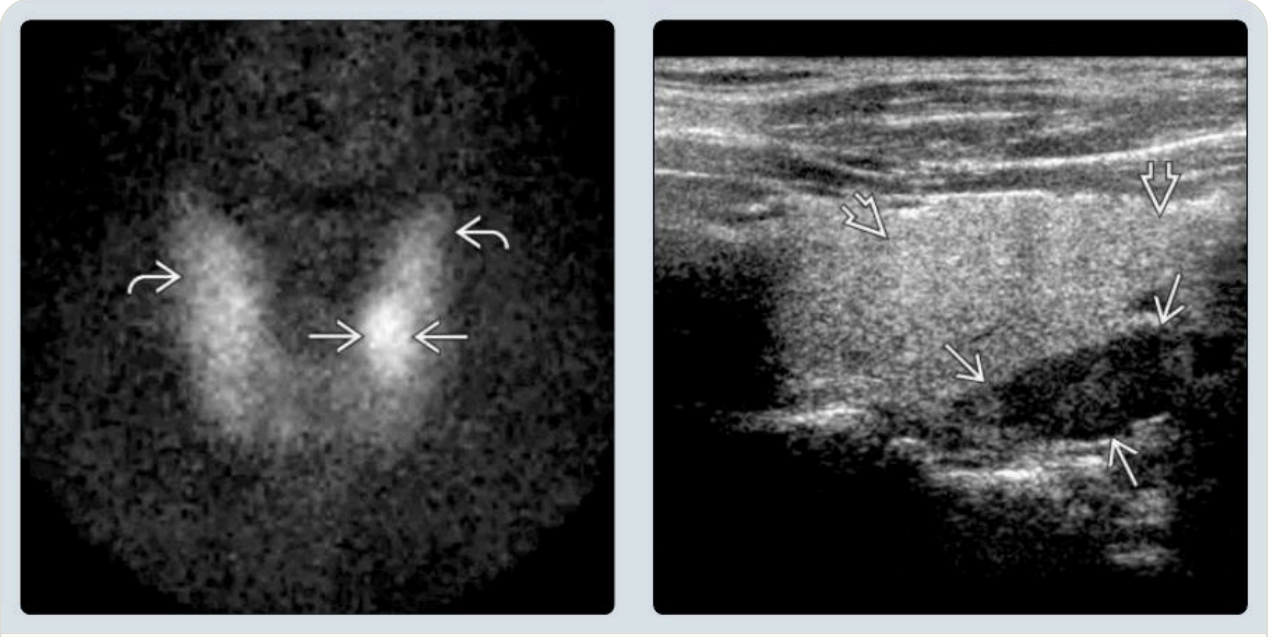


FIGURE 5 Parathyroid adenoma. Left: delayed sestamibi scan, persistent uptake below the left lobe. Right: ultrasound, a hypoechoic oval nodule behind the lower left lobe.
Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 234

2 CHAPTER TWO The parotid

Superficial, deep and the nerve.

2.1 The facial nerve plane

The facial nerve cannot be seen on routine imaging within the parotid, so its plane is inferred. The **retromandibular vein** runs just medial to the nerve, and the nerve lies lateral to it; a tumour lateral to the vein is in the superficial lobe, one medial to it in the deep lobe. A **deep lobe** tumour passes through the **stylomandibular gap** (between the mandibular ramus and the styloid process and its ligament) and pushes the parapharyngeal fat medially. Ultrasound shows the superficial lobe well but not the deep lobe, which is shadowed by the mandible.

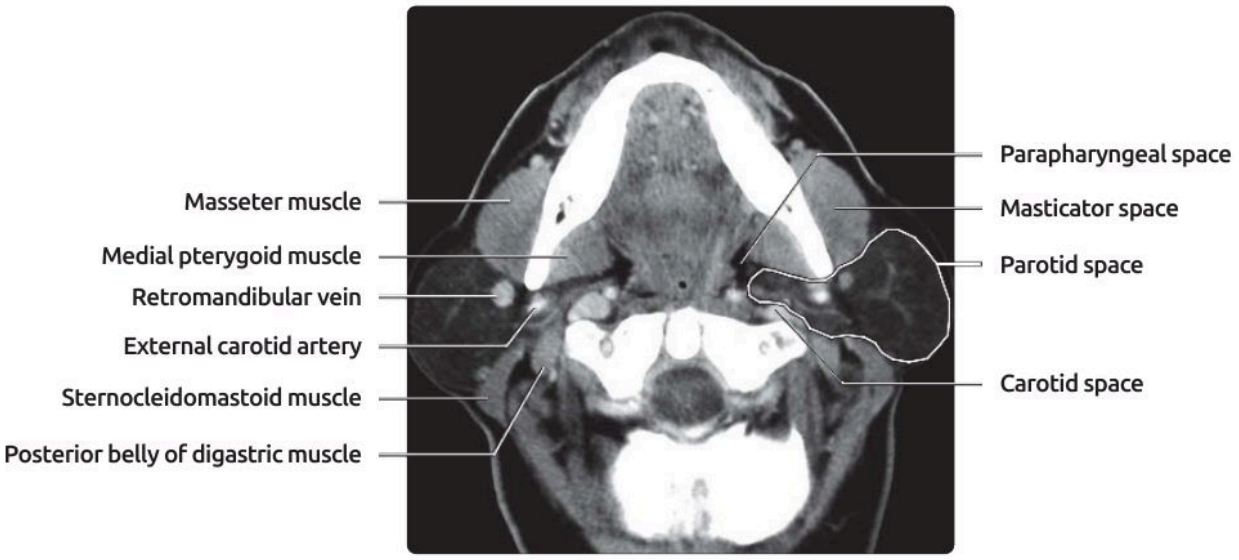


FIGURE 6 Axial contrast CT at the mid oropharynx: the parotid space outlined, with the retromandibular vein lateral to the external carotid artery. The facial nerve runs just lateral to the vein.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 326

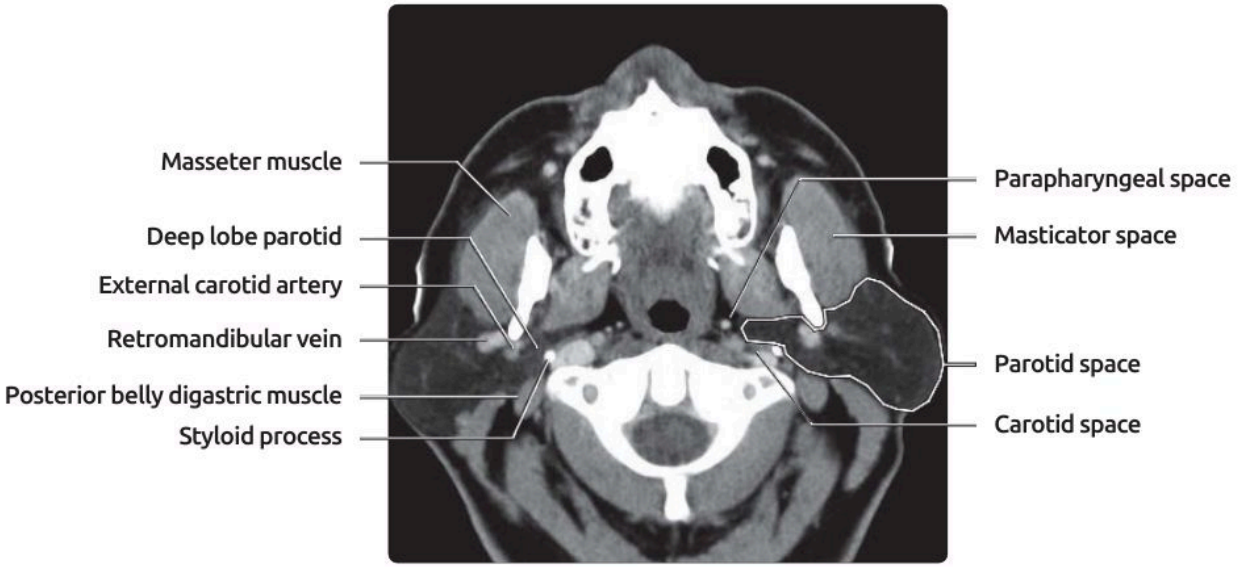


FIGURE 7 The deep lobe of the parotid projecting through the stylomandibular gap to touch the parapharyngeal space.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 325

2.2 Benign tumours

A **pleomorphic adenoma** is well defined, lobulated and typically **very bright on T2**, with a capsule; it is the commonest salivary tumour and can arise anywhere in the

gland or its deep lobe. A **Warthin tumour** occurs in older men who smoke, in the **tail** of the parotid, is often **multiple or bilateral**, may be partly cystic, and is avid on PET.



FIGURE 8 Pleomorphic adenoma. Left: drawing of a small superficial lobe tumour. Right: fat-suppressed T2, a sharply marginated, very bright mass in the superficial lobe, close to the plane of the facial nerve.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 88

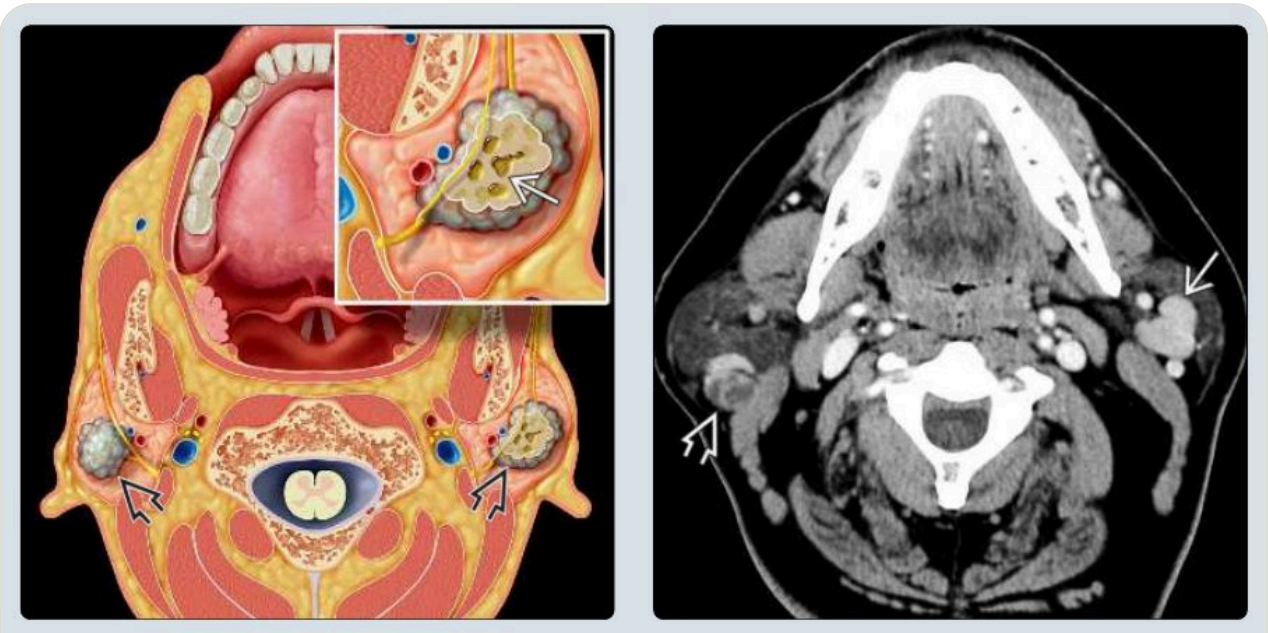


FIGURE 9 Warthin tumours. Left: drawing of bilateral tumours in the parotid tails. Right: contrast CT, a tumour in each parotid.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 92

2.3 Features of malignancy

Ill-defined or infiltrative margins, **low or intermediate T2 signal**, low ADC on diffusion, invasion of the masseter, skin or deep lobe, and abnormal nodes suggest malignancy, although a small low-grade carcinoma can look benign and FNA or core biopsy is still needed. Facial nerve weakness or pain with a parotid mass suggests **perineural spread**: follow the nerve up through the stylomastoid foramen and the mastoid segment of the facial canal on post-gadolinium MRI, and look for loss of the fat in the stylomastoid foramen.

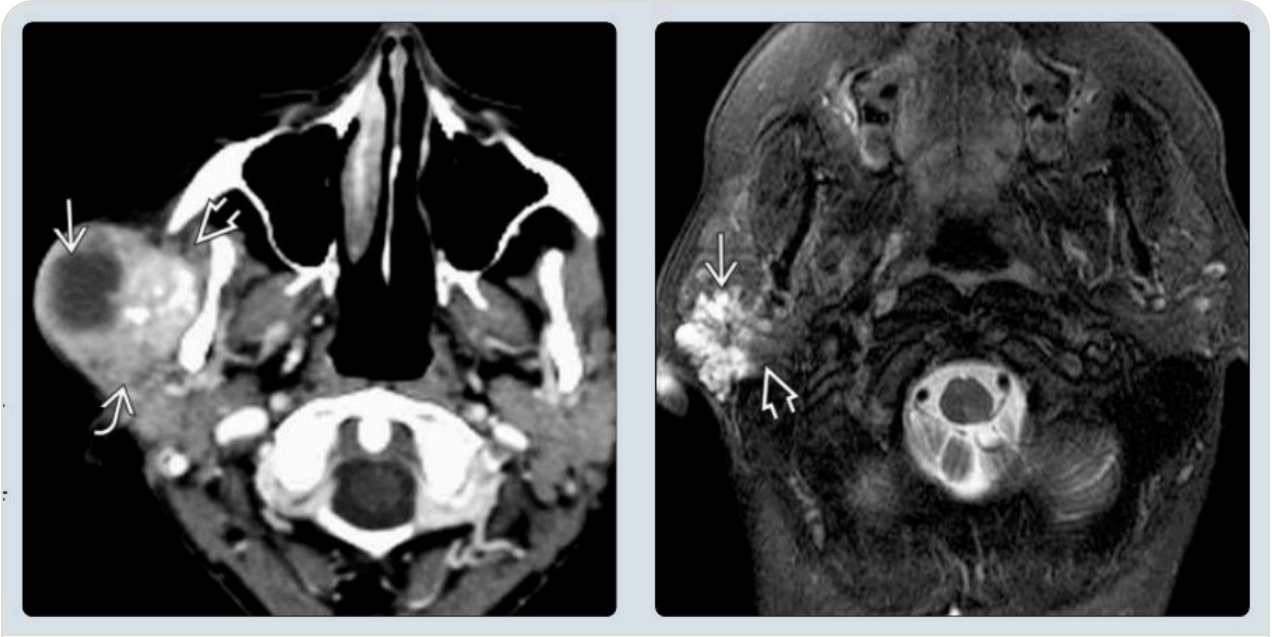


FIGURE 10 Mucoepidermoid carcinoma. Left: contrast CT, a large heterogeneous mass with cystic parts and ill-defined borders invading the masseter. Right: T2, irregular mixed signal with a low-signal solid part.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 98

2.4 Infection, stones and Sjögren's

Sialolithiasis is shown by non-contrast CT or ultrasound; most stones are in the submandibular duct. Acute **parotitis** enlarges and enhances the gland, and contrast CT shows an abscess. **Sjögren's syndrome** gives enlarged glands with many small cysts or a speckled "honeycomb" pattern on ultrasound and MRI; a new dominant solid mass in such a gland raises the question of lymphoma.

3 CHAPTER THREE
The submandibular gland

Gland or node?

3.1 Telling them apart

Masses in the submandibular space are either in the gland or in a level IB node next to it. A mass arising in the gland is surrounded by a "claw" of gland tissue; a node lies outside the gland, usually **in front of the facial vein**, whereas gland masses lie behind it. Submandibular gland tumours are more often malignant than parotid tumours, so a solid mass in the gland needs sampling.

READING A GLAND ULTRASOUND OR SCAN

1. Which gland and where
   - → thyroid lobe and pole
   - → parotid superficial or deep to the retromandibular vein
   - → submandibular gland or node
2. Lesion
   - → composition
   - → echogenicity or T2 signal
   - → shape (taller than wide)
   - → margins
   - → calcification
   - → vascularity

3 **Grade**

- → U or TI-RADS for thyroid
- → benign versus malignant features for salivary

4 **Beyond the gland**

- → extrathyroidal extension
- → trachea, larynx, oesophagus, recurrent nerve
- → masseter, skin, facial nerve and stylomastoid foramen

5 **Nodes**

- → levels III, IV, VI for thyroid
- → intraparotid and level II for parotid

6 **Next step**

- → FNA or core biopsy
- → CT or MRI for retrosternal, deep or invasive disease

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Ultrasound first** for thyroid, salivary and superficial lumps, with FNA.
- **Suspicious thyroid features:** solid, hypoechoic, taller than wide, irregular, microcalcification.
- **U grading and TI-RADS**, and when to sample.

- **CT for retrosternal goitre** and invasive thyroid cancer; the contrast and radioactive iodine issue.
- **Parathyroid localisation:** ultrasound, sestamibi, 4D-CT.
- **Facial nerve plane** and the retromandibular vein; the deep lobe and the stylomandibular gap.
- **Pleomorphic adenoma versus Warthin tumour,** and the features of malignancy.

4.2 Common pitfalls

COMMON PITFALLS

- Ordering CT for a thyroid nodule instead of ultrasound.
- Giving iodinated contrast casually to a patient about to have radioactive iodine.
- Calling a deep lobe parotid tumour "parapharyngeal".
- Missing perineural spread to the stylomastoid foramen in a parotid cancer.
- Mistaking a level IB node for a submandibular gland tumour.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), visceral and parotid space chapters, Imaging Anatomy: Head and Neck (Chapman and Koontz), parotid and visceral spaces, ESR Modern Radiology, Head and Neck Imaging, Dunnebier, Imaging for Otolaryngologists. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.